

## 8 Finance

### 8.5 Payday Loans and Credit Cards

#### 8.5.1 Credit Cards

**Example 8.5.1** Credit cards usually compound interest daily. Suppose that you owe \$1000 on a credit card with a 25% APR. You make a payment of \$100 (after interest is compounded for the month). How much will you still owe the bank? (Assume each month has exactly 30 days)

Use the compound interest formula, where the exponent is just 30. This might take some explaining.

**Example 8.5.2** Suppose that you owe \$1000 on a credit card with 28% APR, and the minimum payment each month is \$50, additionally assume that you make no additional purchases (not a very realistic assumption, I know, but let's make this simple for ourselves). How much will we owe the bank after 3 months?

**Example 8.5.3** Let's go the computer for this one. In the above situation from the previous example, how long will it take us to pay off the bank? How much do we pay?

### 8.5.2 Payday Loans

**Example 8.5.4** Payday loans are short-term loans where the interest is calculated up front. Consider the following example. You need \$400 dollars to make rent, but don't get paid for another two weeks. You decide to take out a payday loan, and the lender charges \$15 dollars interest for every \$100 dollars borrowed. How much interest will you pay if you pay back the loan on time? What is the APR of this loan?